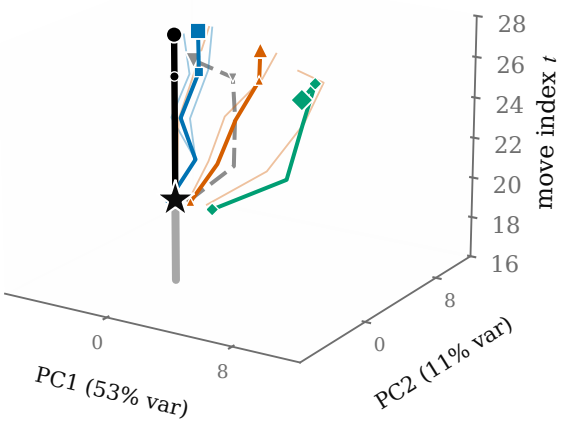
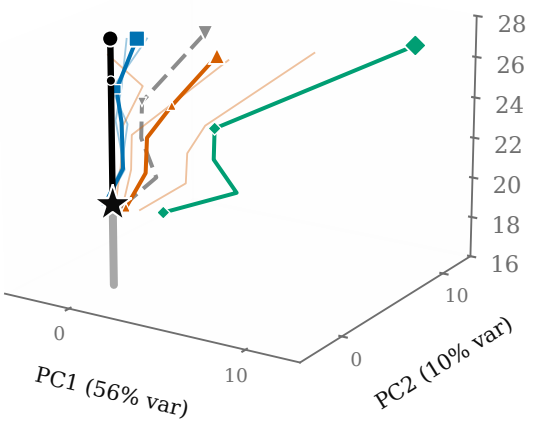
(a) where the branches go: the fork in each representation space, one path per rule

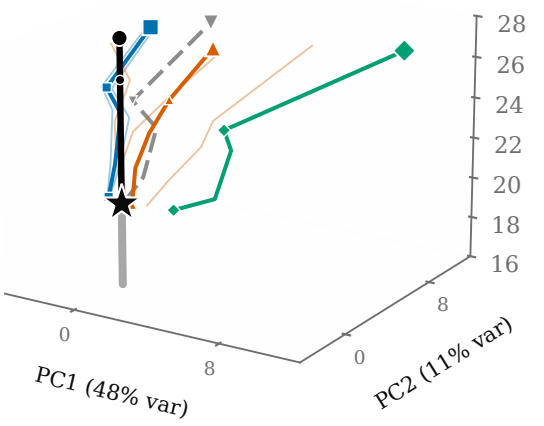
raw layer 5 (reference space)



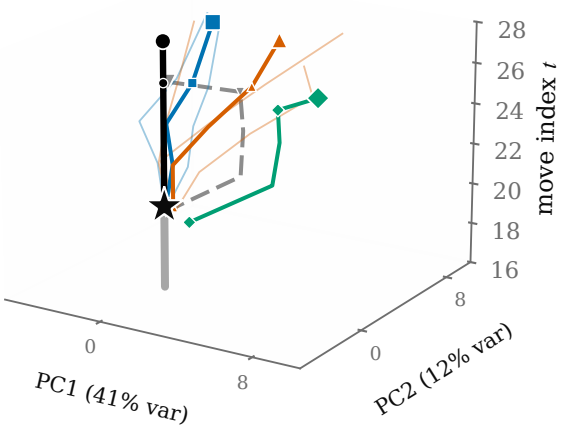
PAnneal (p-anneal, pen 0.042)



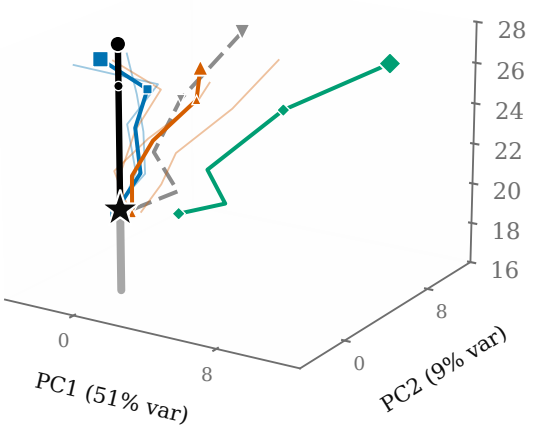
Standard (L1 0.059)



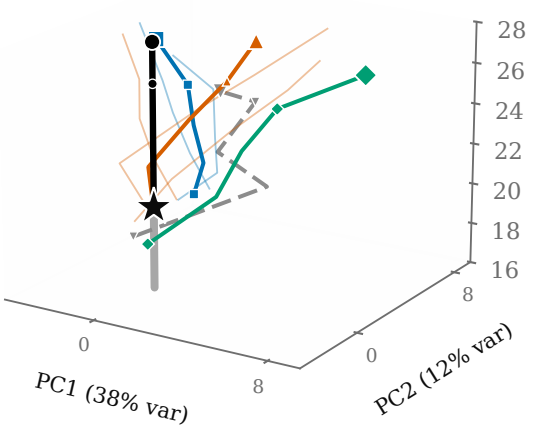
GatedSAE (L1 1.37)



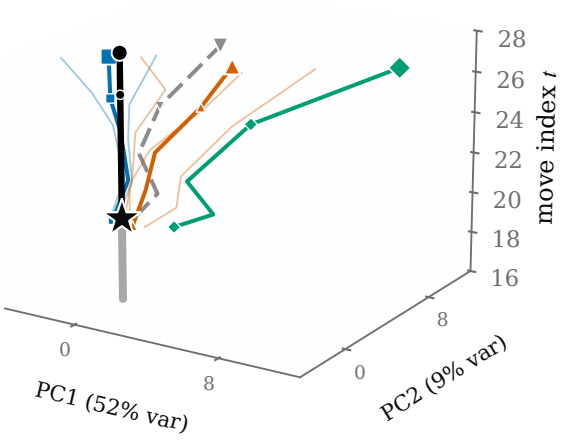
TopK (k=80)



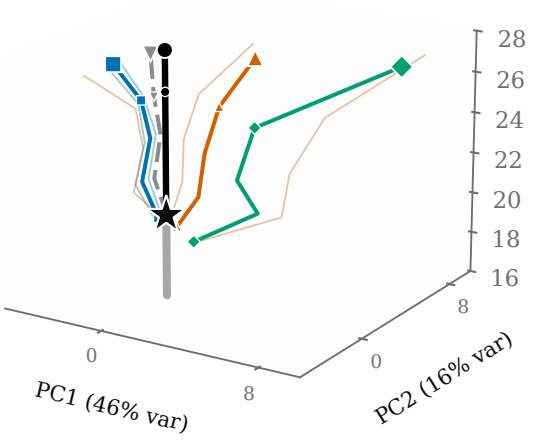
GatedAnneal (p-anneal, pen 0.955)



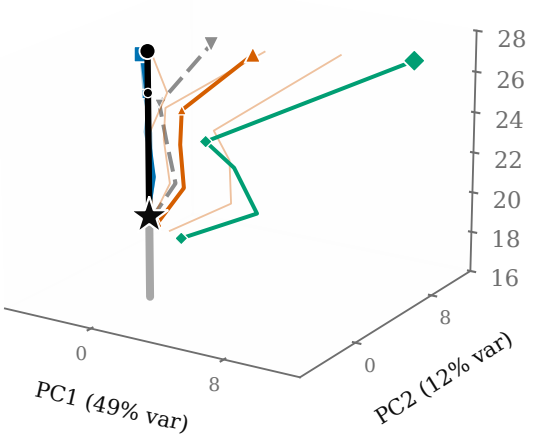
BatchTopK (k=80)



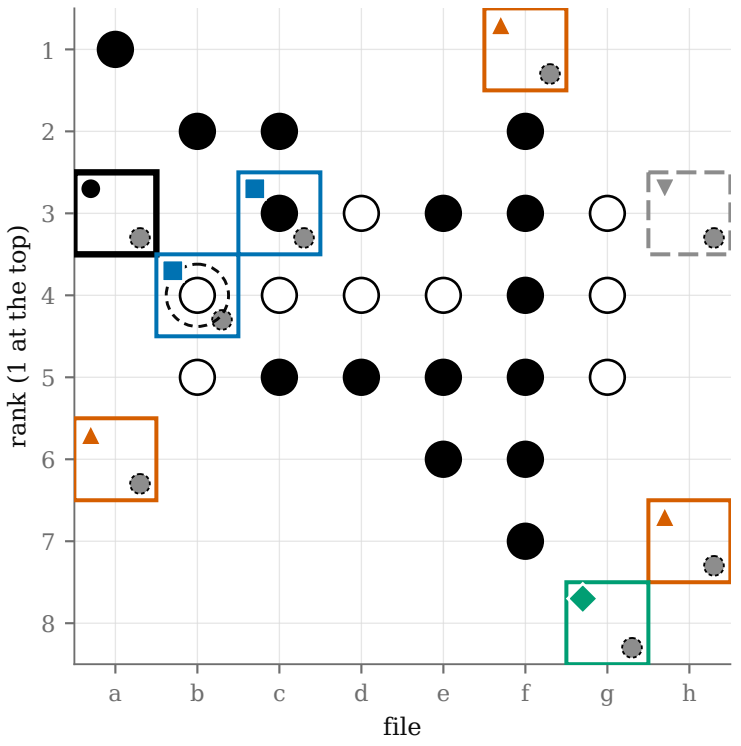
CosineSpaDE (γ=1.0)



SpaDE (λ=1e-3, γ=1e-4)



(b) what they are: the position at t_0 (Black to move), and the move each branch plays there



shared prefix (branches identical)
reference (the game as played)
R1 placement (mean of 2)
R2 bracketing (mean of 3)
R3 turn (mean of 1)
one category inside a rule (within-rule spread)
branch point t_0
control: another legal move (null)

discs in absolute colour, black to move at t_0; outlined cell = a branch square, its marker in the corner so a disc underneath stays visible; small dashed disc in the opposite corner = the disc that branch would place, never a disc on the board; dashed ring = a disc the played move flips; a rule's member categories share one colour and marker here, and are told apart by their square

rollout variant: every branch, the reference included, samples its own continuation from OthelloGPT. Each path is the mean over the branch's K = 4 rollouts, paired on the sample index and taken after φ, which is nonlinear.
Drawn to r = +8: further out illegal and control δ agree to within 0.5% and the fork carries no separation (README).

R1 placement (2): square holds the mover's disc · square holds the opponent's disc
R2 bracketing (3): empty, touches no discs · empty, touches only the mover's discs · empty, touches opponent discs, no line closes
R3 turn (1): flips only for the opponent (turn confusion)